

Supplementary Material

Per-instance Comparison with Published IJSP Solvers on All 70 Taillard Instances

Companion to: “Neighbourhood Structures and Ranking Operators
for the Interval Job Shop Scheduling Problem”

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Description

This document reports per-instance RE (%) and median runtime for all 70 Taillard Interval Job Shop Scheduling Problem (IJSP) instances, comparing four solvers: *fEABC* [1], *ESABC* [2], *TS-N₂* (this work), and *TS-N₈* (this work).

Metric. $RE(\%) = (E[C_{\max}] - LB) / LB \times 100$, where LB is the published lower bound of the *crisp* (deterministic) counterpart of each Taillard instance [3]. The interval IJSP and the crisp JSP are structurally distinct optimisation problems, so these crisp lower bounds are used here only as a common, hardware-independent reference yardstick; they are *indicative* for the interval setting and do not constitute a strict optimality gap. In particular, a negative RE (%) is possible when the interval-makespan midpoint falls below the crisp lower bound, and does not signal a violation.

Columns. For each solver: *Best* = best RE (%) over 30 runs; *Avg* = mean RE (%) over 30 runs (bold = row minimum); *t* = median runtime in seconds (TS) or average runtime in seconds (ABC methods).

Instances. The 70 Taillard IJSP instances (TA1–TA70) are derived from the classical Taillard JSP benchmarks [3] by applying a $\pm 7.5\%$ symmetric interval perturbation to each nominal processing time: $p_{ij} = [0.925 p_{ij}^*, 1.075 p_{ij}^*]$. Seven size classes are covered: 15×15 (TA1–10), 20×15 (TA11–20), 20×20 (TA21–30), 30×15 (TA31–40), 30×20 (TA41–50), 50×15 (TA51–60), 50×20 (TA61–70).

TS configuration. Irace-tuned tabu search (Phase B of the main paper), 30 independent runs per instance, 15-minute wall-clock time limit per run, on an Intel Xeon E5-2680 v4 at 2.40 GHz (WSL2 Linux).

Irace training instances. The irace tuning of Phase B (Section 7.1 of the main paper) used a stratified random subset of 20 of the 82 benchmark instances ($\approx 24\%$), drawn uniformly at random within each size stratum. The 62 instances *not* included in the list below were held out from tuning and contribute the unbiased evaluation reported in the main paper and in Tables 1–2.

- *Classical* (3 of 12): **ft10**, **abz7**, **la27**.
- *Taillard* 15×15 (2 of 10): TA3, TA8.
- *Taillard* 20×15 (2 of 10): TA12, TA17.
- *Taillard* 20×20 (3 of 10): TA21, TA24, TA29.
- *Taillard* 30×15 (2 of 10): TA32, TA36.
- *Taillard* 30×20 (3 of 10): TA43, TA47, TA50.
- *Taillard* 50×15 (2 of 10): TA52, TA57.
- *Taillard* 50×20 (3 of 10): TA63, TA67, TA70.

The full irace input files (`instances.txt`, `parameters.txt`, `scenario.txt`) and the resulting `iracedump.rda` files for each of the five neighbourhoods are archived at [4].

Table 1: Taillard IJSP instances, Part I (TA1–TA40). $RE(\%) = (E[\mathbf{C}_{\max}] - LB) / LB \times 100$. All methods: 30 runs. Bold: row-minimum average RE. t = median runtime (s) for TS; average for ABC methods.

15×15	TA1	1231	4.71	6.73	3.7	4.51	7.27	5.4	1.26	1.85	20	1.26	2.09	34
	TA2	1244	2.65	4.20	3.9	3.26	4.87	5.2	0.24	1.65	19	0.88	1.57	37
	TA3	1218	3.65	5.63	4.4	3.74	5.57	5.7	1.35	2.34	23	1.35	2.28	35
	TA4	1175	4.38	6.99	4.0	5.15	6.96	5.9	0.34	1.31	21	0.34	1.06	37
	TA5	1224	3.19	4.89	3.5	2.49	4.27	5.0	1.06	2.22	19	1.06	1.94	33
	TA6	1238	3.39	5.61	4.2	2.34	4.89	5.5	1.29	1.97	20	1.62	1.92	33
	TA7	1227	2.81	4.86	3.6	2.57	4.42	5.3	0.61	1.77	16	0.61	1.79	27
	TA8	1217	2.55	5.41	3.9	1.97	5.68	5.2	1.15	2.37	17	1.15	2.18	31
	TA9	1274	4.47	8.25	4.1	5.46	8.26	5.7	2.35	2.73	17	2.35	2.74	30
	TA10	1241	3.14	6.16	3.8	3.71	6.44	5.3	0.24	2.23	19	0.24	1.90	31
20×15	TA11	1357	6.26	9.37	6.6	6.74	8.37	9.7	2.76	4.11	45	2.76	4.30	65
	TA12	1367	4.46	5.99	5.8	2.93	5.28	8.5	1.57	2.21	31	1.65	2.25	59
	TA13	1342	6.26	8.88	7.1	7.19	9.25	9.8	2.38	3.91	43	2.01	4.00	68
	TA14	1345	3.75	5.02	6.3	2.49	4.43	8.6	0.45	1.82	35	0.74	1.95	55
	TA15	1339	5.94	8.53	6.3	4.59	7.05	9.6	2.43	3.96	44	2.80	4.18	65
	TA16	1360	7.02	8.98	6.3	6.84	9.22	8.6	1.95	3.28	38	2.10	3.23	60
	TA17	1462	5.75	8.72	6.1	4.31	8.06	8.0	1.92	3.14	40	2.02	3.21	58
	TA18	1377	9.40	11.79	6.6	9.37	11.63	9.6	3.70	5.23	44	3.96	5.87	75
	TA19	1332	8.52	11.36	5.7	7.96	9.70	8.9	1.99	4.54	48	1.91	4.38	86
	TA20	1348	4.97	8.09	6.2	5.27	7.02	10.2	2.15	3.54	41	1.97	3.51	70
20×20	TA21	1642	6.52	8.07	6.9	5.42	7.47	11.2	2.38	3.75	60	2.74	3.90	97
	TA22	1561	8.65	10.53	8.1	8.23	9.77	11.8	4.16	5.88	59	4.80	6.00	91
	TA23	1518	10.01	11.56	8.0	9.49	11.23	11.8	4.45	5.80	49	4.41	6.26	85
	TA24	1644	5.14	7.47	7.8	5.29	6.91	11.0	1.43	2.86	58	1.34	2.80	101
	TA25	1558	9.27	10.97	7.8	8.06	11.22	11.2	5.10	6.50	59	5.30	6.81	97
	TA26	1591	10.50	13.05	8.4	9.99	12.93	11.7	6.66	8.02	54	6.03	8.05	89
	TA27	1652	8.57	10.42	8.3	7.45	9.82	11.6	3.93	5.08	58	3.63	5.03	95
	TA28	1603	6.46	8.21	7.5	6.24	7.52	10.9	2.18	3.27	60	2.25	3.38	99
	TA29	1583	7.99	10.21	7.6	7.64	9.63	12.0	4.36	5.10	50	4.14	4.88	82
	TA30	1528	9.75	12.22	7.7	10.01	12.04	10.6	5.99	7.13	70	5.82	7.06	102
30×15	TA31	1764	6.01	8.78	12.5	5.70	7.81	18.7	1.02	1.97	102	0.54	2.17	171
	TA32	1774	10.68	13.07	13.4	9.53	11.84	19.8	4.17	5.32	127	3.83	5.55	211
	TA33	1788	9.40	11.71	13.3	9.03	10.58	18.8	3.75	4.61	106	3.16	4.77	172
	TA34	1828	8.23	9.63	12.2	7.28	8.76	17.3	3.39	4.73	98	3.04	4.71	135
	TA35	2007	1.47	3.25	11.7	1.59	3.92	13.0	0.00	0.26	61	0.00	0.27	94
	TA36	1819	7.94	9.63	12.7	6.24	8.83	18.2	1.29	2.82	105	1.18	2.87	174
	TA37	1771	8.16	9.70	12.7	7.06	9.11	16.4	2.88	3.85	113	3.16	4.06	192
	TA38	1673	8.52	10.65	12.1	7.29	9.51	18.9	3.17	4.35	122	2.81	4.48	193
	TA39	1795	6.13	8.35	13.2	5.46	7.82	18.0	1.45	2.67	90	1.67	2.60	142
	TA40	1651	11.18	13.42	13.1	11.21	12.71	17.8	4.18	6.17	102	4.82	6.57	153

Table 2: Taillard IJSP instances, Part II (TA41–TA70). $RE(\%) = (E[C_{\max}] - LB) / LB \times 100$. All methods: 30 runs. Bold: row-minimum average RE. t = median runtime (s) for TS; average for ABC methods.

30×20	TA41	1906	17.58	20.55	16.1	16.34	19.24	22.4	9.68	12.12	158	10.36	12.31	257
	TA42	1884	14.15	17.45	15.3	14.46	16.86	20.8	6.58	8.36	192	6.53	8.77	323
	TA43	1809	16.06	18.13	16.5	13.85	16.74	24.2	5.67	8.72	197	6.77	8.79	294
	TA44	1948	11.88	14.41	16.8	10.47	12.78	25.9	4.72	6.86	151	5.78	7.12	233
	TA45	1997	6.99	9.45	17.0	6.76	8.40	24.2	2.70	4.17	132	2.88	4.37	220
	TA46	1957	13.00	15.91	18.2	12.31	14.87	26.2	6.87	8.69	168	5.93	8.70	276
	TA47	1807	16.55	20.44	16.3	15.33	18.64	24.0	9.91	11.53	161	10.29	12.20	274
	TA48	1912	11.90	13.90	15.1	11.30	13.88	22.1	6.15	7.87	161	6.22	7.91	267
	TA49	1931	11.08	13.19	17.1	10.56	12.98	21.3	5.00	6.49	155	4.79	6.64	230
	TA50	1833	16.56	20.36	16.3	16.50	18.83	22.1	9.22	11.43	163	10.37	11.66	264
50×15	TA51	2760	5.51	7.20	27.3	3.73	5.76	40.6	0.00	0.02	204	0.00	0.09	291
	TA52	2756	4.03	5.93	27.5	2.87	4.47	43.2	0.00	0.10	212	0.00	0.10	272
	TA53	2717	3.66	5.40	27.5	2.82	4.04	37.9	0.04	0.04	162	0.04	0.05	233
	TA54	2839	0.63	2.96	24.3	0.33	1.75	34.3	0.00	0.02	108	0.00	0.04	158
	TA55	2679	6.01	8.44	29.3	5.88	7.52	39.2	0.34	0.87	244	0.34	0.88	407
	TA56	2781	3.25	5.04	23.7	3.00	4.57	36.9	0.11	0.27	156	0.11	0.33	258
	TA57	2943	1.63	3.61	26.9	1.22	2.48	41.1	0.00	0.03	127	0.00	0.00	197
	TA58	2885	3.08	4.94	26.7	1.94	3.60	38.4	0.69	0.69	125	0.69	0.69	184
	TA59	2655	5.63	7.50	27.8	4.73	6.62	38.9	0.45	0.75	209	0.45	0.81	385
	TA60	2723	4.43	6.07	24.7	3.64	4.67	34.9	0.11	0.41	185	0.11	0.35	261
50×20	TA61	2868	6.76	8.84	33.5	5.81	7.77	49.2	0.47	1.83	542	0.47	2.18	784
	TA62	2869	9.41	11.74	33.7	8.66	10.40	44.7	3.66	5.02	460	3.31	5.08	653
	TA63	2755	7.64	9.19	30.6	6.35	7.87	45.8	0.45	1.69	545	0.42	1.66	750
	TA64	2702	6.94	8.36	33.6	5.94	6.78	47.0	0.59	1.50	521	0.13	1.44	689
	TA65	2725	8.42	9.98	33.8	7.30	9.05	47.5	1.19	2.12	574	1.08	2.32	741
	TA66	2845	6.80	8.56	32.1	6.05	7.55	42.8	1.20	2.05	527	0.93	1.95	655
	TA67	2825	6.50	8.95	34.0	6.50	8.24	45.4	1.08	2.30	585	1.31	2.24	788
	TA68	2784	6.66	8.01	32.7	4.87	6.85	46.6	0.29	1.10	421	0.27	1.15	534
	TA69	3071	5.08	6.59	29.6	3.70	5.90	45.0	0.00	0.12	440	0.00	0.38	461
	TA70	2995	8.43	9.91	30.9	7.20	8.72	42.9	1.59	2.53	664	1.80	2.90	701

References

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